

Jawaharlal Nehru Govt. Engineering College Sundernagar, Mandi (H.P.)

Course Name: Introduction to Machine Learning Course Code: CSPC-513 Examination: 1st Periodical Examination

Scheme: NEP Branch: CSE (AI & ML) Semester: V

Question Paper for 1st Periodical Examination (September 2025) (Session July-Dec 2025)

Time: One and a half hours

Max. Marks: 30

Date: 03.10.2025

Note: Attempt all questions.

Course Outcomes:

CO1. Design and implement machine learning solutions to classification, regression and clustering problems.

CO2. Evaluate and interpret the results of the different ML techniques.

CO3. Design and implement various machine learning algorithms in a range of real-world applications.

Q.1 Answer the following questions:

(1x5=5)

- (a) Define PTE of machine learning.
- (b) What is the need of data splitting in machine learning?
- (c) List two differences between numeric and ordinal data.
- (d) Give examples of independent and dependent features.
- (e) Which transformation is used for right skewed distribution where most values are large? How does it remove skewness?

Q.2 Explain any five data encoding schemes.

(5)

Q.3(a) Differentiate Min-Max and Z Score normalization with example.

(2)

Q.3(b) Write any three cross validation techniques.

(3)

Q.4 A company wants to predict monthly electricity bills for households based on the factors: size of the house, number of occupants, number of appliances. Answer the following:

(1+4)

- (a) Which ML algorithm will be suitable for this task?
- (b) A household has a house size of 1600 sq.ft, 5 occupants, and 11 appliances. Predict the electricity bill using a regression model. Use the following dataset:

House Size (sq.ft)	No. of Occupants	No. of Appliances	Monthly Electricity Bill (Rs.)
800	2	5	1200
950	3	6	1500
1200	4	8	2100
1500	5	10	2600
1800	4	12	3000
2000	6	14	3500
2200	5	15	3900
2500	7	18	4500

Q.5 Attempt any two.

(5x2=10)

- (a) Explain data preprocessing process with the help of an example.
- (b) Explain a probabilistic classification algorithm with the help of an example.
- (c) Consider the dataset below. Find whether the student with MST1 = 8, MST2 = 13, Attendance = 76, Assignment = 5 passes or fails.

Student	MST1	MST2	Attendance (%)	Assignment	Result
S1	18	20	85	12	Pass
S2	10	25	90	14	Fail

Student	MST1	MST2	Attendance (%)	Assignment	Result
S3	15	8	80	10	Fail
S4	22	24	60	9	Fail
S5	16	18	92	6	Pass
S6	14	12	70	8	Fail
S7	28	26	95	15	Pass
S8	12	14	75	5	Pass
S9	20	19	78	4	Fail
S10	11	13	82	7	Fail

KL: Knowledge Level: L1 - Remember, L2 - Understand, L3 - Apply, L4 - Analyze, L5 - Evaluate, L6 - Create

Course Advisor: Shivani (Assistant Professor, CSE)